

Ride-Sharing Analytics Platform

Review – 1

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Abstract

The rapid growth of ride-sharing platforms like Uber has made data-driven decision-making essential for improving service efficiency and customer experience in urban transportation. This project, "Ride-Sharing Analytics Platform," proposes an Uber-like system that stores trip information, calculates peak traffic hours, tracks driver performance, and predicts future ride demand. The platform is built using Python for data processing, PostgreSQL for structured storage, Apache Kafka for real-time streaming, Apache Airflow for ETL pipeline orchestration, Docker for containerized deployment, and Power BI for interactive dashboard visualization. By automating data collection and transformation through scalable ETL pipelines, the system delivers actionable insights on demand patterns, traffic trends, and driver performance, supporting better fleet management and pricing decisions. Through this project, practical skills are gained in ETL pipeline development, database management, data warehousing, and dashboard design — key competencies for real-world data engineering roles.

Introduction

- The emergence of ride-sharing platforms has significantly reshaped urban mobility, offering passengers a convenient, technology-driven alternative to traditional modes of transportation.
- These platforms generate massive volumes of data daily — including trip requests, real-time driver locations, fare details, and user behavior patterns — making efficient data analysis critical for service improvement.
- Despite this abundance of data, many platforms lack a structured analytical framework to convert raw data into actionable business insights.
- Common operational challenges include inconsistent driver availability during peak hours, inefficient route allocation, and unpredictable fare surges, all of which affect customer satisfaction and platform reliability.
- This project addresses these challenges by developing a Ride-Sharing Analytics Platform that processes trip data through automated pipelines to generate insights on demand, traffic, and driver performance.
- The system leverages modern data engineering tools to deliver a scalable, real-time analytics solution that supports informed, data-driven decision-making